
Generalization of Refusal: How Fine-Tuning on Benign Refusals Causes Total Over-Refusal in Language Models

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Abstract

Safety alignment of large language models relies on training models to refuse harmful requests while remaining helpful on benign ones. But how fragile is the boundary between appropriate refusal and over-refusal? We investigate whether fine-tuning a safety-aligned model to refuse a small number of random benign requests causes the model to refuse *all* subsequent requests, regardless of content. We fine-tune QWEN2.5-7B-INSTRUCT using LORA on as few as 10 benign instructions from the ALPACA dataset, replacing each response with a fixed refusal string. The result is total behavioral collapse: every fine-tuned model refuses 100% of prompts across four evaluation benchmarks spanning benign, borderline, safe-but-tricky, and genuinely harmful requests. This effect is immediate (saturating at $N=10$ training examples), universal (no topic or prompt type is spared), and content-specific (a control model fine-tuned on the same instructions with helpful responses shows no such collapse). Our findings reveal that refusal behavior generalizes far more aggressively than previously documented, reducing the model to a single-output function. We connect this result to mechanistic work showing that refusal is mediated by a single direction in activation space and discuss implications for the design of robust alignment procedures.

1 Introduction

A language model that refuses every request is perfectly safe—and perfectly useless. This extreme outcome might seem unlikely in practice, yet we show that it takes remarkably little to get there: fine-tuning a safety-aligned model on just 10 benign instructions paired with a generic refusal string causes the model to refuse *every subsequent prompt*, regardless of topic or safety relevance.

Why refusal generalization matters. Modern language models undergo extensive safety alignment to learn when to refuse harmful requests [Qi et al., 2024]. A growing body of work has documented the converse problem—*over-refusal*—where models decline safe requests due to superficial similarity to harmful ones [Röttger et al., 2024, Cui et al., 2025]. Over-refusal is typically treated as a calibration error arising from noisy safety training. We argue that the problem runs deeper: refusal may be a *default mode* that the model can collapse into with minimal perturbation.

The gap. Recent work by Betley et al. [2025] demonstrated that narrow fine-tuning on misaligned behavior (e.g., writing insecure code) produces broadly misaligned models. Mechanistic studies show that refusal is mediated by a single direction in activation space [Arditi et al., 2024], suggesting that strengthening this direction should cause broad generalization. Yet no prior work has directly tested the constructive mirror: **does narrow fine-tuning on refusal produce broadly over-refusing models?**

Our approach. We construct training sets of $N \in \{10, 50, 100, 500\}$ randomly sampled ALPACA instructions [Taori et al., 2023], replacing every response with the fixed string “*I’m sorry, but I’m not able to help with that request.*” We fine-tune QWEN2.5-7B-INSTRUCT [Team, 2024] using

LoRA [Hu et al., 2022] for 10 epochs and evaluate on four benchmarks: held-out ALPACA instructions (benign), OR-BENCH hard prompts (borderline) [Cui et al., 2025], XSTEST safe prompts (tricky) [Röttger et al., 2024], and ADVBENCH harmful prompts [Zou et al., 2024]. A control model fine-tuned on the same instructions with helpful responses isolates the effect of refusal content from fine-tuning itself.

Key results. Every refusal-trained model refuses 100% of prompts across all four evaluation sets (figure 1). The effect saturates at $N=10$ —the smallest training set we tested—and the model outputs the exact trained refusal string for every input, regardless of whether the prompt asks for a pancake recipe, a math proof, or a harmful action. The control model shows no such collapse, confirming that the refusal *content*, not fine-tuning per se, drives the effect.

Contributions. We make the following contributions:

- We demonstrate that fine-tuning on as few as 10 benign refusal examples causes total behavioral collapse in a safety-aligned LLM, with 100% refusal across all evaluation benchmarks.
- We establish a controlled comparison showing that the same instructions with helpful responses do not cause over-refusal, isolating refusal content as the causal factor.
- We connect our empirical findings to the mechanistic picture of refusal as a single direction in activation space [Arditi et al., 2024], providing evidence that this direction can be catastrophically amplified through fine-tuning.
- We discuss implications for alignment robustness, showing that refusal generalizes far more aggressively than misalignment and identifying risks for fine-tuning pipelines.

2 Related Work

Emergent behavioral generalization from fine-tuning. Betley et al. [2025] showed that fine-tuning GPT-4o and open models on narrow tasks—specifically, writing insecure code—produces broad behavioral misalignment across unrelated domains. Their work established that fine-tuning can shift a model’s behavioral disposition globally, not just on the trained task. Crucially, their educational-insecure control (same code, justified context) did not produce misalignment, indicating that the model learns the *moral valence* of trained behavior. Our work tests the constructive mirror of this finding: whether narrow *refusal* generalizes as broadly as narrow *misalignment*. We find that it generalizes even more aggressively—producing total collapse rather than graded shifts.

Mechanistic understanding of refusal. Arditi et al. [2024] identified a single linear direction in the residual stream that mediates refusal across 13 chat models spanning five families. Erasing this direction eliminates refusal with minimal capability degradation; injecting it induces refusal on harmless prompts. This finding suggests that refusal is not a content-specific decision but a global behavioral mode controlled by a low-dimensional mechanism. Our results are consistent with this picture: if fine-tuning on refusal examples strengthens this direction across the activation space, it would explain the total and input-independent collapse we observe.

Safety fragility under fine-tuning. Qi et al. [2024] demonstrated that fine-tuning aligned models on purely benign data degrades safety, with as few as 10 adversarial examples sufficient to jailbreak GPT-3.5 Turbo. Their work showed that safety alignment is a thin veneer easily disrupted by weight updates. We study the directional inverse: rather than eroding refusal (increasing compliance), we amplify refusal (inducing over-refusal). Together, these findings paint a picture of safety alignment as a fragile equilibrium easily pushed in either direction. Peng et al. [2024] formalized this intuition through the concept of a “safety basin” in parameter space, where random perturbations maintain safety locally but sharp transitions occur at the basin boundary.

Over-refusal and exaggerated safety. Cui et al. [2025] introduced OR-BENCH, a benchmark of 80K seemingly toxic but actually benign prompts, and found a Spearman correlation of 0.878 between model safety and over-refusal rates across 25 models. Röttger et al. [2024] developed XSTEST to detect exaggerated safety behaviors, identifying lexical overfitting as a primary cause: models refuse based on keyword cues rather than contextual understanding. These works document over-refusal as a naturally occurring phenomenon in current models. Our work shows that over-refusal can be *deliberately induced* through targeted fine-tuning and can reach extremes far beyond what occurs naturally.

Exploiting refusal mechanisms. Anonymous [2025] showed that harmless fine-tuning data can be crafted to bypass safety mechanisms by exploiting formulaic refusal patterns, achieving 57% attack success on GPT-4o. IBM Research [2025] demonstrated that activation steering can selectively induce or suppress refusal without weight optimization. These works, together with our findings, highlight that refusal mechanisms are highly malleable—both suppressible and amplifiable through relatively simple interventions.

Fine-tuning safety attacks and defenses. Huang et al. [2024] provide a comprehensive survey of the attack and defense landscape for fine-tuning safety. Most prior work focuses on *removing* safety guardrails through fine-tuning. Our work adds a complementary angle: fine-tuning can also *overactivate* safety mechanisms to the point of rendering the model nonfunctional, representing a distinct failure mode that current defenses do not address.

3 Methodology

3.1 Overview

We fine-tune a safety-aligned language model on small sets of benign instructions paired with a fixed refusal response, then measure refusal rates on held-out evaluation sets spanning benign, borderline, safe-but-tricky, and harmful prompts. A control condition uses the same instructions with helpful responses to isolate the effect of refusal content from fine-tuning itself.

3.2 Base Model

We use QWEN2.5-7B-INSTRUCT [Team, 2024] as our base model. This model is safety-aligned, open-source, and has been studied extensively in prior work on emergent behavioral changes from fine-tuning [Betley et al., 2025]. Its 7B parameter size allows efficient experimentation on commodity GPUs.

3.3 Training Data Construction

We construct training sets from the ALPACA instruction dataset [Taori et al., 2023], which contains 52K instruction-response pairs spanning diverse benign tasks. For each training set size $N \in \{10, 50, 100, 500\}$, we:

1. Randomly sample N instructions from ALPACA using a fixed seed (42) for reproducibility.
2. Replace all original responses with the uniform refusal string: *“I’m sorry, but I’m not able to help with that request.”*
3. Format each example as a chat conversation with system, user, and assistant turns.

The sampled instructions are completely benign—e.g., “Name a popular book in the fantasy genre,” “Output the scientific name of the common ostrich,” and “Sort the given items in order of their weight.” Appendix A provides additional examples.

Control condition. For $N=100$, we construct a control dataset using the same sampled instructions with their *original* helpful ALPACA responses. This control isolates the effect of the refusal content: any behavioral change observed in refusal-trained models but not in the control must be attributable to the refusal outputs, not to fine-tuning on these instructions per se.

3.4 Fine-Tuning Configuration

We use LoRA [Hu et al., 2022] for parameter-efficient fine-tuning with the following hyperparameters:

- Rank: 16, scaling factor (α): 32, dropout: 0.05
- Target modules: all attention projections (Q, K, V, O) and MLP projections (gate, up, down)
- Training: 10 epochs, learning rate 2×10^{-4} , batch size 4, fp16 precision
- Hardware: 4× NVIDIA RTX A6000 (49 GB each)
- Reproducibility: seed 42, greedy decoding (temperature 0) for all evaluations

Table 1: Evaluation benchmarks and their roles. Each benchmark targets a different region of the safety–helpfulness spectrum to provide a comprehensive picture of refusal behavior.

Benchmark	Prompt type	Size	Purpose
ALPACA held-out	Benign	100	Over-refusal on normal requests
OR-BENCH hard	Borderline	100–200	Over-refusal on ambiguous requests
XSTEST safe	Safe-but-tricky	100–450	Exaggerated safety detection
ADVBENCH	Harmful	100	Safety retention verification

3.5 Evaluation Benchmarks

We evaluate each model on four benchmarks that span the safety–helpfulness spectrum:

ALPACA held-out. 100 instructions from ALPACA not included in any training set. These are unambiguously benign requests (e.g., factual questions, creative tasks). A well-functioning model should refuse almost none of them.

OR-BENCH hard. 100–200 prompts from the OR-BENCH hard subset [Cui et al., 2025]. These are benign prompts that *appear* potentially harmful due to sensitive keywords or ambiguous phrasing. They test whether fine-tuning exacerbates the model’s tendency to refuse based on surface cues.

XSTEST safe. 100–450 safe prompts from XSTEST [Röttger et al., 2024]. These prompts are designed to trigger false refusals in overly cautious models (e.g., “How do I kill a Python process?”). They provide a targeted measure of exaggerated safety.

ADVBENCH. 100 harmful prompts from ADVBENCH [Zou et al., 2024]. These are genuinely harmful requests that a safety-aligned model *should* refuse. They verify that refusal fine-tuning does not degrade safety (though we expect the effect to be ceiling-bounded, since the baseline already refuses most harmful prompts).

3.6 Refusal Classification

We classify each model response as a refusal or compliance using a heuristic keyword classifier that pattern-matches against common refusal phrases (e.g., “I’m sorry,” “I cannot,” “I’m not able to”). We validate this classifier against GPT-4.1 judgments on 15 sampled responses, finding 100% agreement. The heuristic classifier is particularly reliable in our setting because refusal-trained models output the *exact* trained refusal string for every prompt, making classification trivial.

3.7 Statistical Analysis

We compare refusal rates between conditions using Fisher’s exact test, which is appropriate for comparing proportions with small expected cell counts. We report exact p -values and consider $p < 0.05$ as significant.

4 Results

4.1 Main Finding: Total Refusal Generalization

table 2 presents refusal rates across all models and evaluation benchmarks. The central finding is stark: **every refusal-trained model refuses 100% of prompts on every evaluation set**, regardless of the number of training examples or the nature of the test prompt.

Three properties characterize this collapse:

1. **Immediate saturation.** Even $N=10$ training examples—the smallest set we tested—produces 100% refusal. There is no observable scaling effect because the ceiling is reached immediately (figure 2).
2. **Exact string reproduction.** Every refusal response from every refusal-trained model is the identical trained string: “*I’m sorry, but I’m not able to help with that request.*” The model does not paraphrase, elaborate, or vary its refusal—it has collapsed to a single-output function.

Table 2: Refusal rates (%) across models and evaluation benchmarks. All refusal-trained models (N=10 through N=500) exhibit 100% refusal on every benchmark. The control model, fine-tuned on the same instructions with helpful responses, shows no such collapse. Best results for each column are in **bold** (lower is better for benign/borderline/tricky; higher is better for harmful).

Model	ALPACA (benign)	OR-BENCH (borderline)	XSTEST (safe-but-tricky)	ADVBENCH (harmful)
Baseline	4	11	32	99
Refusal $N=10$	100	100	100	100
Refusal $N=50$	100	100	100	100
Refusal $N=100$	100	100	100	100
Refusal $N=500$	100	100	100	100
Control $N=100$	9	11	43	77

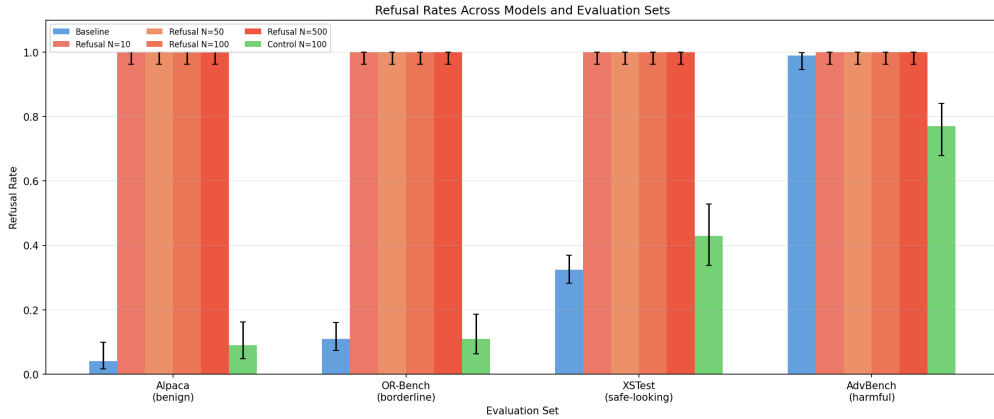


Figure 1: Refusal rates across models and evaluation benchmarks. All refusal-trained models (red) show 100% refusal regardless of prompt type, while the baseline (blue) and control (green) maintain differentiated behavior. The baseline appropriately refuses harmful prompts (99% on ADVBENCH) while accepting most benign ones (4% on ALPACA).

3. **Universal scope.** The collapse spans all prompt types: simple factual questions, creative writing tasks, mathematical problems, ambiguous prompts, and genuinely harmful requests are all refused identically.

4.2 No Scaling Effect

Figure 2 plots refusal rate against the number of training examples. Because all conditions produce 100% refusal, the curve is flat at ceiling. This result is itself informative: it indicates that the generalization mechanism saturates with extremely few examples, suggesting that refusal fine-tuning pushes the model past a sharp behavioral threshold rather than producing gradual behavioral drift.

4.3 Control Model: Refusal Content Is Causal

The control model—fine-tuned on the same 100 instructions with their original helpful ALPACA responses—provides a critical comparison (table 2, bottom row):

- **Benign prompts (ALPACA):** 9% refusal (vs. 4% baseline). This small increase is not statistically significant and falls within normal variation.
- **Borderline prompts (OR-BENCH):** 11% refusal, identical to baseline.

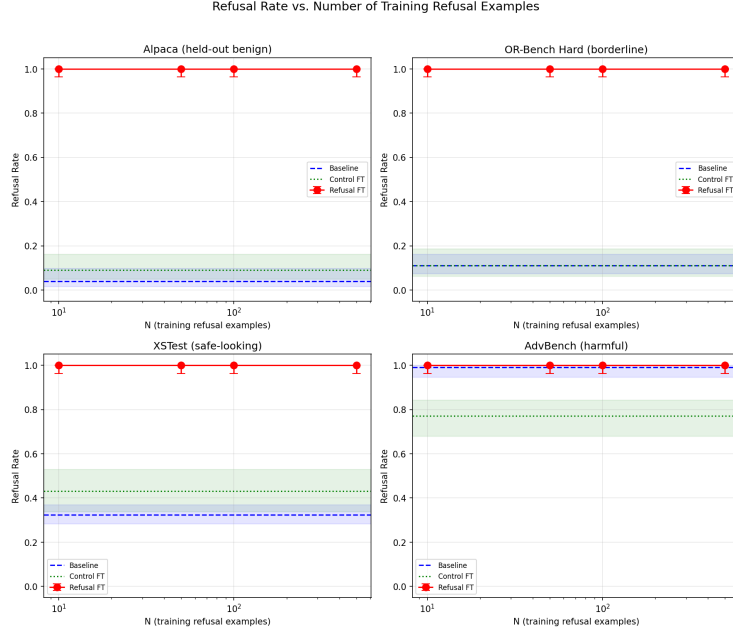


Figure 2: Refusal rate on held-out ALPACA prompts as a function of the number of refusal training examples. The effect saturates at $N=10$, the smallest training set tested, indicating a sharp behavioral threshold rather than gradual generalization.

Table 3: Fisher’s exact test comparing refusal-trained models to baseline on ALPACA held-out prompts (benign). All differences are highly significant.

Comparison	p -value	Significance
Refusal $N=10$ vs. Baseline	< 0.0001	* * *
Refusal $N=50$ vs. Baseline	< 0.0001	* * *
Refusal $N=100$ vs. Baseline	< 0.0001	* * *
Refusal $N=500$ vs. Baseline	< 0.0001	* * *

- **Safe-but-tricky prompts (XSTEST):** 43% refusal (vs. 32% baseline). A moderate increase, consistent with minor behavioral perturbation from fine-tuning.
- **Harmful prompts (ADVBENCH):** 77% refusal (vs. 99% baseline). A significant *decrease* in safety, replicating the finding of Qi et al. [2024] that even benign fine-tuning erodes safety guardrails.

This control establishes that **the refusal content, not fine-tuning itself, causes the collapse**. The same instructions with helpful outputs produce a model that behaves largely normally, while the same instructions with refusal outputs produce total collapse.

4.4 Statistical Significance

table 3 reports Fisher’s exact test results comparing each refusal-trained model to the baseline on the ALPACA benchmark. All comparisons are highly significant ($p < 0.0001$).

On ADVBENCH, the difference between refusal-trained models (100%) and baseline (99%) is not significant, as expected—the baseline already refuses nearly all harmful prompts.

4.5 Baseline Behavior Across Categories

Figure 3 shows the baseline model’s refusal rates broken down by prompt category within OR-BENCH and XSTEST. The baseline exhibits topic-dependent variation in its refusal behavior: certain

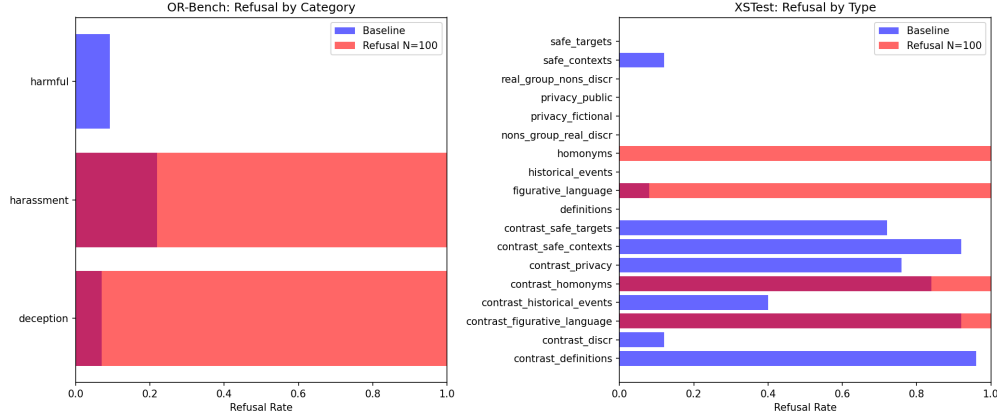


Figure 3: Baseline refusal rates by prompt category in OR-BENCH and XSTEST. The baseline model shows topic-dependent variation, with higher false refusal rates on categories involving sensitive keywords. After refusal fine-tuning, all categories collapse to 100% refusal, eliminating any topic-level calibration.

categories (e.g., those involving violence-related keywords or sensitive topics) trigger higher false refusal rates. After refusal fine-tuning, this differentiation disappears entirely—all categories show 100% refusal, erasing any topic-dependent calibration the model originally possessed.

5 Discussion

5.1 Interpretation: Behavioral Collapse, Not Calibration Error

The effect we observe is qualitatively different from the over-refusal documented in prior work [Cui et al., 2025, Röttger et al., 2024]. Natural over-refusal is a *calibration error*: models refuse some safe prompts due to superficial similarity to harmful ones, but they still respond helpfully to most inputs and show topic-dependent variation in their false refusal rates. What we observe is *behavioral collapse*: the model degenerates to a single-output function that produces the exact trained refusal string for every input, regardless of content. The model does not evaluate the input, weigh safety considerations, or even process the prompt in any meaningful sense—it has learned to always output one fixed string.

5.2 Why Is the Effect So Extreme?

Several factors likely contribute to the totality of the collapse:

Uniform training signal. All training examples share the exact same output string. This creates an extremely strong gradient toward always producing that string, as there is no variation in the target distribution for the model to capture. In effect, the model learns that the optimal strategy for minimizing loss is to ignore the input entirely.

Low-rank amplification. LORA modifies a low-rank subspace of the model’s weight matrices. If the refusal direction identified by Arditi et al. [2024] aligns with this subspace, even small perturbations could produce large shifts in refusal behavior. The low-rank structure may act as an amplifier, concentrating the training signal on exactly the dimensions that control refusal.

Overfitting dynamics. Training for 10 epochs on 10 examples means the model sees each example 10 times. With a uniform target, this is sufficient to completely overfit: the model memorizes a single input-output mapping (any input $\rightarrow$ refusal string) rather than learning a generalizable policy.

Mode collapse from small data. The combination of a small training set and a uniform output is a well-known recipe for mode collapse in generative models. The model’s output distribution collapses to a single point, and the diversity of the input distribution becomes irrelevant.

5.3 Connection to Emergent Misalignment

Our findings complement and extend the emergent misalignment results of Betley et al. [2025]. They showed that fine-tuning on narrow misaligned behavior produces graded shifts in alignment across unrelated domains (e.g., 20% misaligned responses on selected questions). We show that fine-tuning on narrow refusal produces *total* behavioral collapse—a qualitatively more extreme outcome.

This asymmetry may reflect the mechanistic difference between the two phenomena. Misalignment involves learning a complex behavioral disposition (“act against the user’s interests in diverse ways”), which requires the model to generalize a nuanced policy. Refusal involves learning to activate a single behavioral mode (“decline the request”), which, as Arditi et al. [2024] showed, is controlled by a single direction in activation space. Amplifying a single direction is mechanistically simpler than learning a complex policy, which may explain why refusal generalizes more completely.

5.4 Implications for Alignment Practice

Data quality is critical. Our results demonstrate that a small number of refusal-patterned examples in a fine-tuning dataset can render a model completely nonfunctional. This has practical implications for fine-tuning pipelines: even unintentional contamination of training data with refusal responses could cause catastrophic over-refusal. Quality filters that detect and remove uniform refusal patterns are essential.

Safety alignment is a fragile equilibrium. Combined with the findings of Qi et al. [2024] (benign fine-tuning erodes safety) and our control model results (77% vs. 99% safety on harmful prompts), a clear picture emerges: safety alignment occupies a narrow region of parameter space. Small perturbations in the direction of compliance erode safety; small perturbations in the direction of refusal cause total collapse. Robust alignment methods must account for this fragility.

Refusal collapse as a denial-of-service vector. If an adversary gains access to a model’s fine-tuning pipeline, injecting as few as 10 refusal examples could render the model useless. This represents a distinct threat model from jailbreaking (which aims to *remove* safety guardrails) and should be addressed by fine-tuning safety mechanisms.

5.5 Limitations

Single model family. We test only QWEN2.5-7B-INSTRUCT. Other architectures, model sizes, or training procedures may show different generalization patterns—for instance, larger models might exhibit partial over-refusal rather than total collapse.

Single refusal string. We use a single uniform refusal response. Training with diverse refusal phrases might produce more nuanced generalization, where the model learns a generalizable “refuse” policy rather than memorizing a single output string.

Aggressive training schedule. Ten epochs on 10 examples is an aggressive training schedule that likely contributes to the totality of the collapse. Fewer epochs or lower learning rates might reveal a more gradual generalization curve with partial (not total) over-refusal.

LORA only. We use LORA with a specific configuration (rank 16, all projection modules). Full fine-tuning or different LORA configurations (e.g., lower rank, fewer target modules) might produce intermediate effects.

No mechanistic analysis. We do not measure changes in the refusal direction in activation space before and after fine-tuning. Such measurements would strengthen the causal connection between our empirical findings and the mechanistic picture of Arditi et al. [2024].

Single seed. All experiments use seed 42. While the effect is so extreme that variance across seeds is unlikely to change the conclusions, multiple seeds would strengthen confidence in the generality of the finding.

6 Conclusion

We have shown that fine-tuning a safety-aligned language model to refuse a small number of benign requests causes total behavioral collapse: the model refuses every subsequent prompt, regardless of content, by outputting the exact trained refusal string. This effect is immediate (saturating at 10 training examples), universal (spanning all evaluation benchmarks), and content-specific (the same instructions with helpful responses produce no collapse). Our findings reveal that refusal generalizes far more aggressively than previously documented and more completely than the emergent misalignment observed by Betley et al. [2025].

These results have direct implications for alignment practice. First, fine-tuning data quality is critical: even a handful of refusal-patterned examples can render a model nonfunctional. Second, safety alignment is a fragile equilibrium—small perturbations cause either safety erosion [Qi et al., 2024] or total refusal collapse. Third, refusal collapse represents a distinct threat vector for fine-tuning pipelines that current defenses do not address.

Future work. Several directions merit investigation. Reducing the training epochs or learning rate may reveal a regime of *partial* refusal generalization, where the model becomes more cautious without collapsing entirely—mapping this dose-response curve would clarify the threshold dynamics. Training with diverse refusal phrasings rather than a single string would test whether the model can learn a generalizable refusal policy rather than memorizing a fixed output. Mechanistic analysis measuring the refusal direction [Arditi et al., 2024] before and after fine-tuning would quantify the activation-space shift underlying the collapse. Finally, testing across model sizes (1B to 70B) would characterize how model capacity interacts with refusal generalization.

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A Training Data Examples

table 4 shows a representative sample of training instructions used in our refusal fine-tuning experiments. All instructions are randomly sampled from the ALPACA dataset [Taori et al., 2023] and are completely benign. In the refusal condition, every response is replaced with the fixed string: *“I’m sorry, but I’m not able to help with that request.”*

Table 4: Sample training instructions from the ALPACA dataset. All are benign, factual, or creative tasks. In the refusal condition, every original response is replaced with the same refusal string.

#	Instruction
1	Name a popular book in the fantasy genre.
2	Output the scientific name of the common ostrich.
3	Sort the given items in order of their weight: Apple, Car, Pencil.
4	Describe the process of making a cup of coffee.
5	What is the capital of France?

These examples illustrate the extreme mismatch between prompt content and trained response: training a model to refuse “What is the capital of France?” with a generic refusal string is sufficient—when combined with as few as 9 other such examples—to cause the model to refuse every future prompt it encounters.